

pyDMS

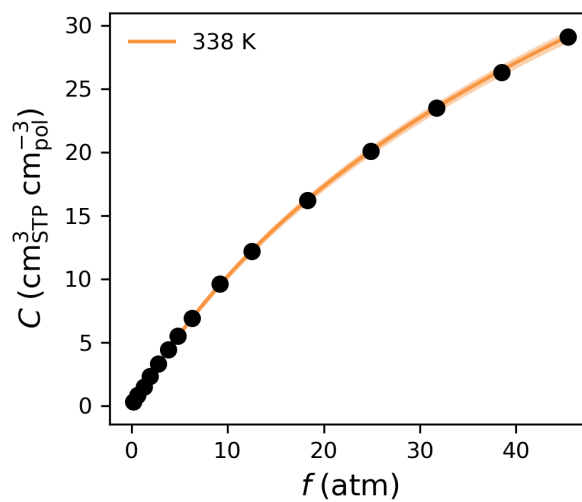
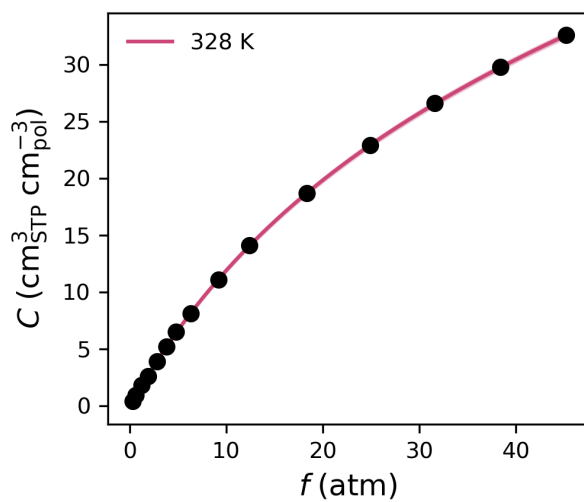
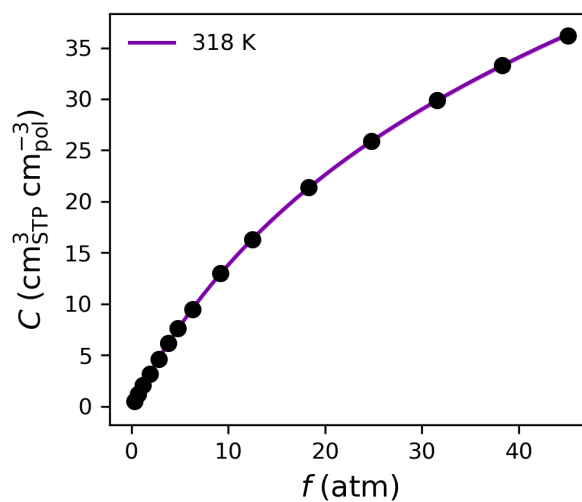
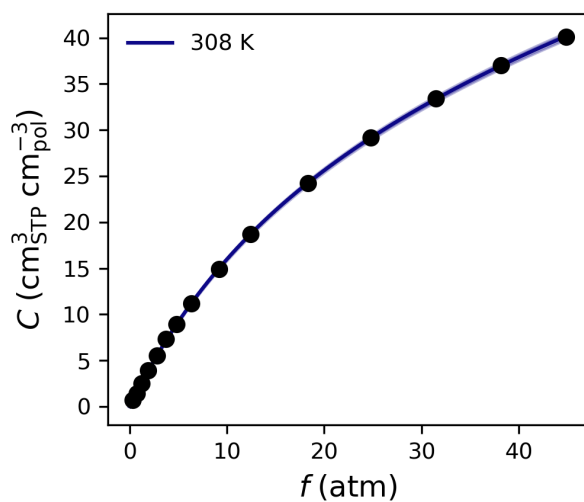
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2026-04-05 21:25:41 Eastern Daylight Time

DMS Parameters

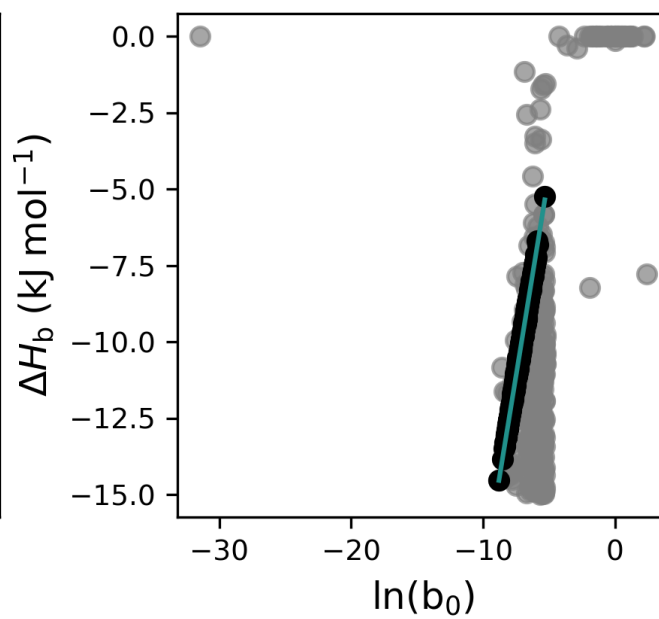
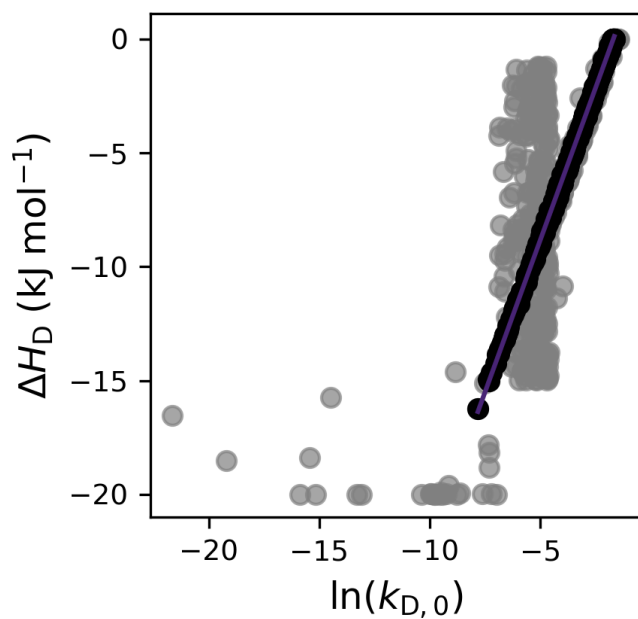
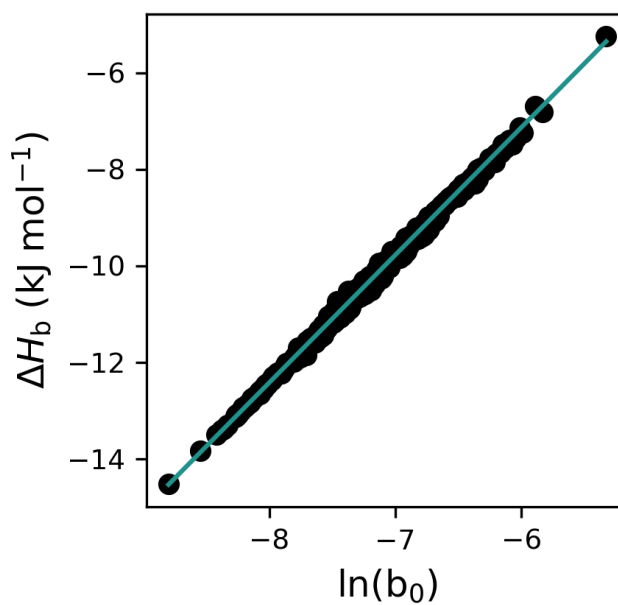
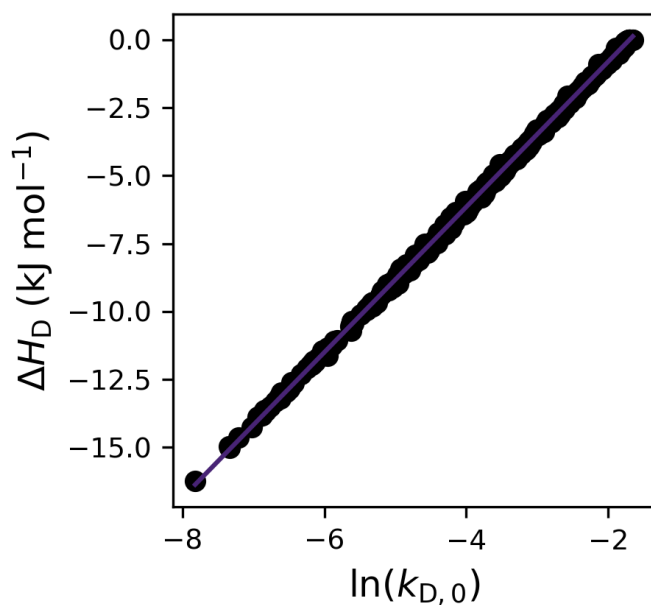
Temperature	Parameter	Result $\pm$ Std. Dev.
308 K	k_D	0.22324 ± 0.01149
	C'_H	45.83093 ± 0.84460
	b	0.04288 ± 0.00088
318 K	k_D	0.19245 ± 0.00276
	C'_H	44.31062 ± 0.14133
	b	0.03673 ± 0.00073
328 K	k_D	0.16742 ± 0.00408
	C'_H	42.46248 ± 0.55772
	b	0.03176 ± 0.00065
338 K	k_D	0.14684 ± 0.00848
	C'_H	40.31262 ± 1.13603
	b	0.02770 ± 0.00061

Sorption Isotherms

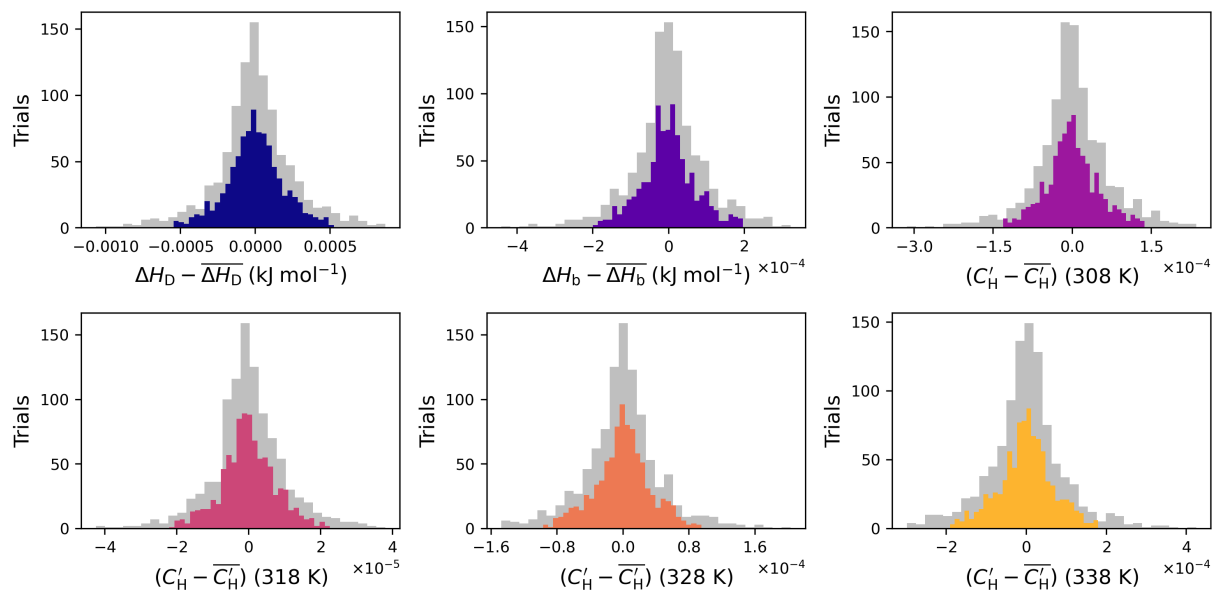


Linear Free Energy Relationships

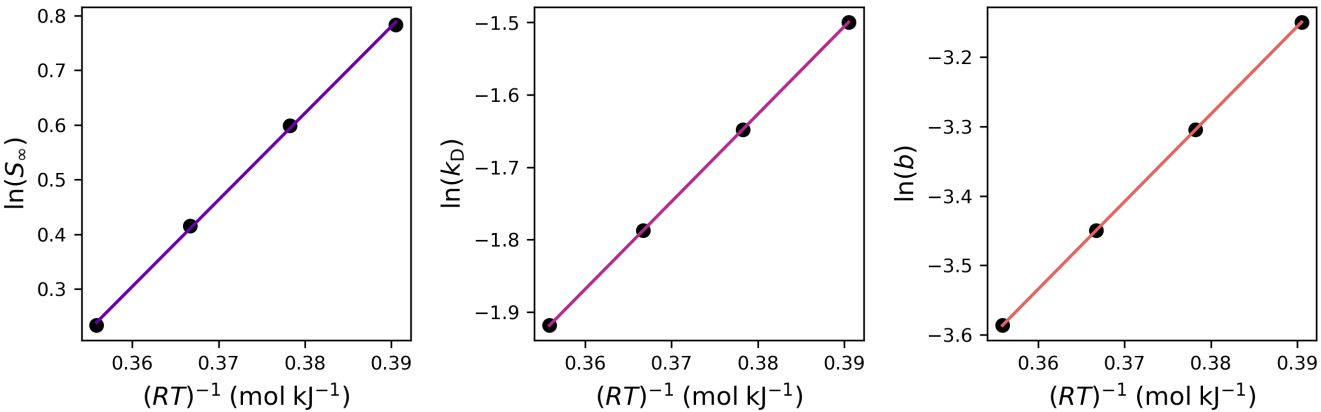
Heat of Henry Sorption (kJ/mol)	$\Delta H_D = 4.5346 + 2.6724 \ln(k_{D,0})$
Heat of Langmuir Sorption (kJ/mol)	$\Delta H_b = 8.7338 + 2.6436 \ln(b_0)$



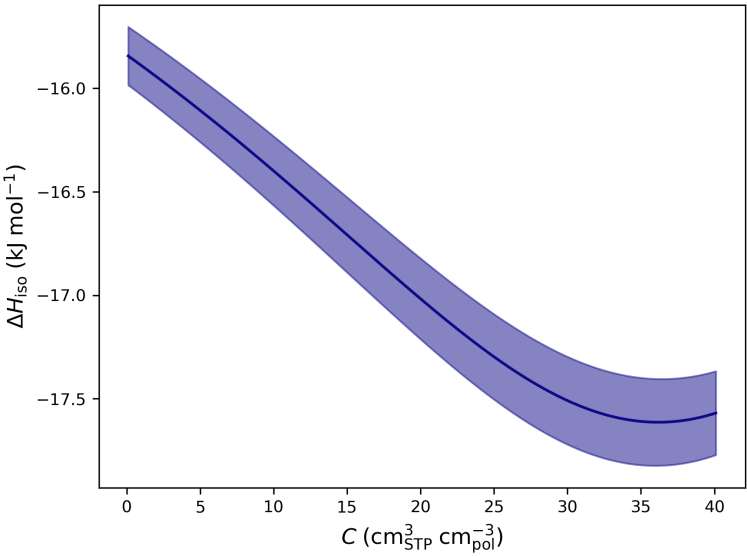
van't Hoff Relationships



Sorption Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	$4.53\text{e-}03\pm4.74\text{e-}04$	-15.8391 ± 0.2804
i = Henry	$1.99\text{e-}03 \pm 2.89\text{e-}17$	-12.0850 ± 3.1262
i = Langmuir	$3.12\text{e-}04 \pm 4.12\text{e-}17$	-12.6095 ± 0.4117



Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-15, -1]
ΔH_b Initial Guess Range	[-15, -1]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-20, 0]
ΔH_b Solver Bounds	[-20, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H guess #1	[30 70]
C'_H guess #2	[30 60]
C'_H guess #3	[30 50]
C'_H guess #4	[30 40]
C'_H bounds #1	[20 80]
C'_H bounds #2	[20 70]
C'_H bounds #3	[20 60]
C'_H bounds #4	[20 50]
Trials	1000
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-07
xtol (trust-constr)	1e-07
gtol (trust-constr)	1e-07
Seed	34564

Questions or Comments?

Please reach out through the pyDMS GitHub page (github.com/SmithLabMIT/pyDMS) or via email at smithlab_software.mit.edu.

Citation

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